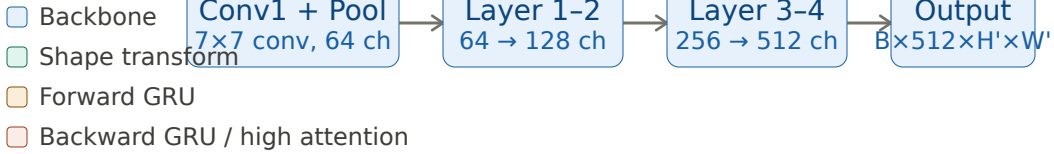


Spectrogram image
 $224 \times 224 \times 3$ (RGB)

ResNet18 backbone (ImageNet pretrained, avgpool + fc removed)



Frequency dimension pooling
AdaptiveAvgPool2d(1, None) → $B \times 512 \times 1 \times W'$

Squeeze + permute
 $B \times W' \times 512$ (W' time steps)

Bidirectional GRU — layer 1

→ Forward GRU (hidden=256)
 $t=1 \rightarrow t=2 \rightarrow \dots \rightarrow t=W'$

← Backward GRU (hidden=256)
 $t=W' \rightarrow \dots \rightarrow t=2 \rightarrow t=1$

Concat → $B \times W' \times 512$

Bidirectional GRU — layer 2
Same structure → $B \times W' \times 512$

Attention pooling

h_1
 $t=1$

...

h_{n-1}
 $t=n-1$

$h_n \star$
 $t=n$

Linear(512→1)
+ Softmax

Weighted sum → $B \times 512$

Classifier head

Linear 512→256

GELU + Drop

Linear 256→N classes

Genre score (logit)